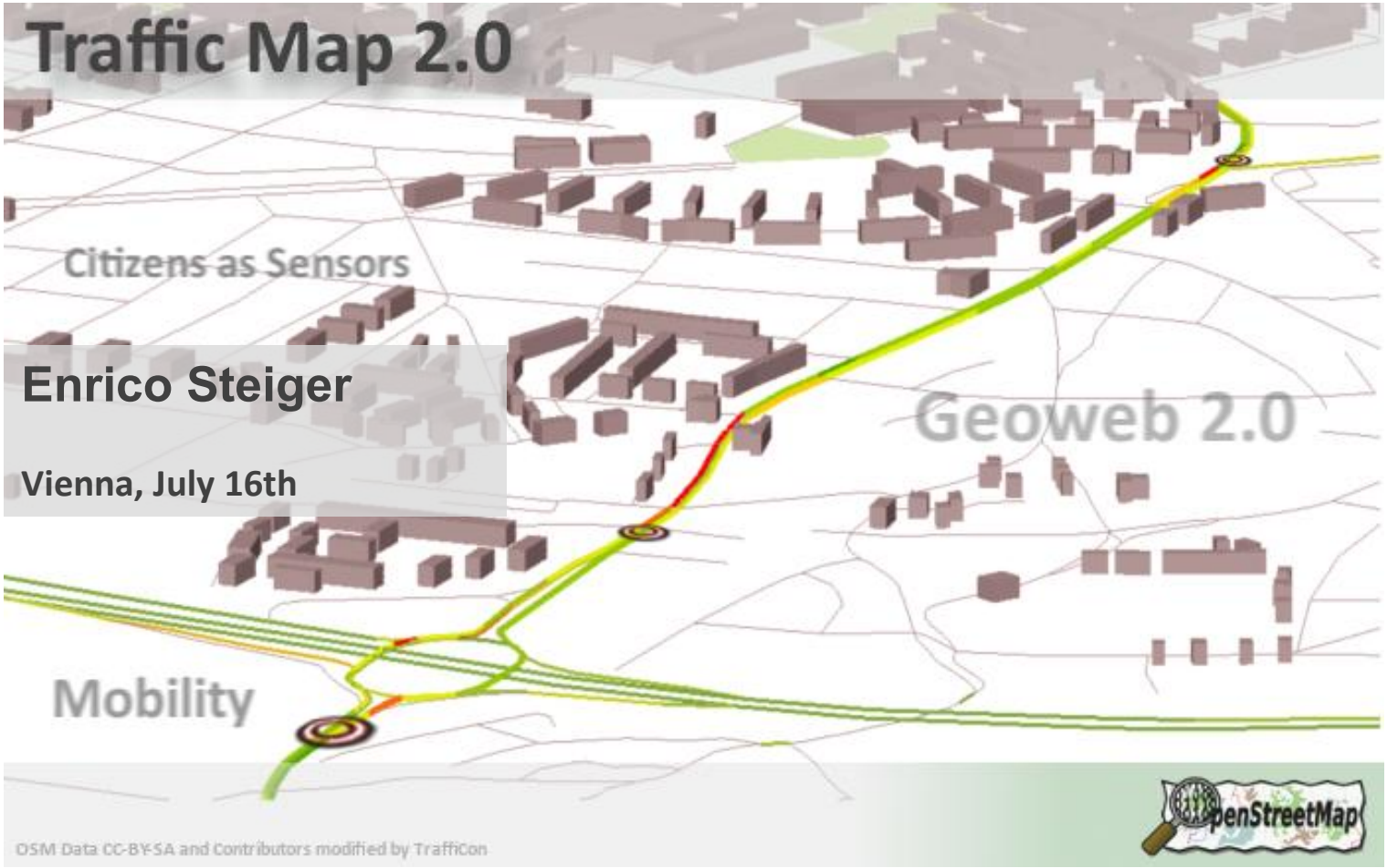


Enrico Steiger

Vienna, July 16th

TraffiCon – Traffic Consultants GmbH
Enrico Steiger
Schillerstraße 25 - 5020 Salzburg, www.trafficon.eu

Research Coperation in Salzburg

rsa iSPACE Research Studios Austria Forschungsgesellschaft mbH

- Spatial Research
- Sensors
- Accessibility models
- Energy
- Potential models
- Indicator
- Public Transport
- web: real-time
- Standards

TraffiCon Verkehrstechnik, Verkehrstelematik

- Transport telematics
- Traffic Models
- Traffic Planning
- Mobility Services
- OpenStreetMap Applications
- ITS
- Public Transport

salzburg research

- Mobile Applications
- Travel Time Models
- Software architecture
- Floating-Car-Data
- Mobile Information Services

Z_GIS

- ITS
- Research



Applied Research and development in the area of transport telematics and geoinformatics

Mobility – quo vadis?

- Number of journeys are steadily increasing
- One consequence are highly frequented roads.
- According to calculations by the Federal Office of Statistics every German spends about 70 hours annually in traffic jams.
- Economy loses [illegible]
- Forecast Salzburg 2015: Travel times increases by 15 Minutes / Trip

Goal is a detection of bottlenecks
inside transport networks as well as a
temporally, spatially and modal shift of trips.

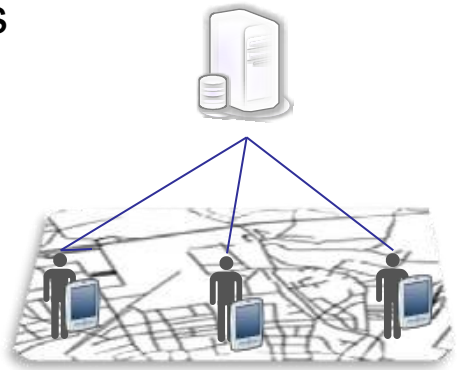






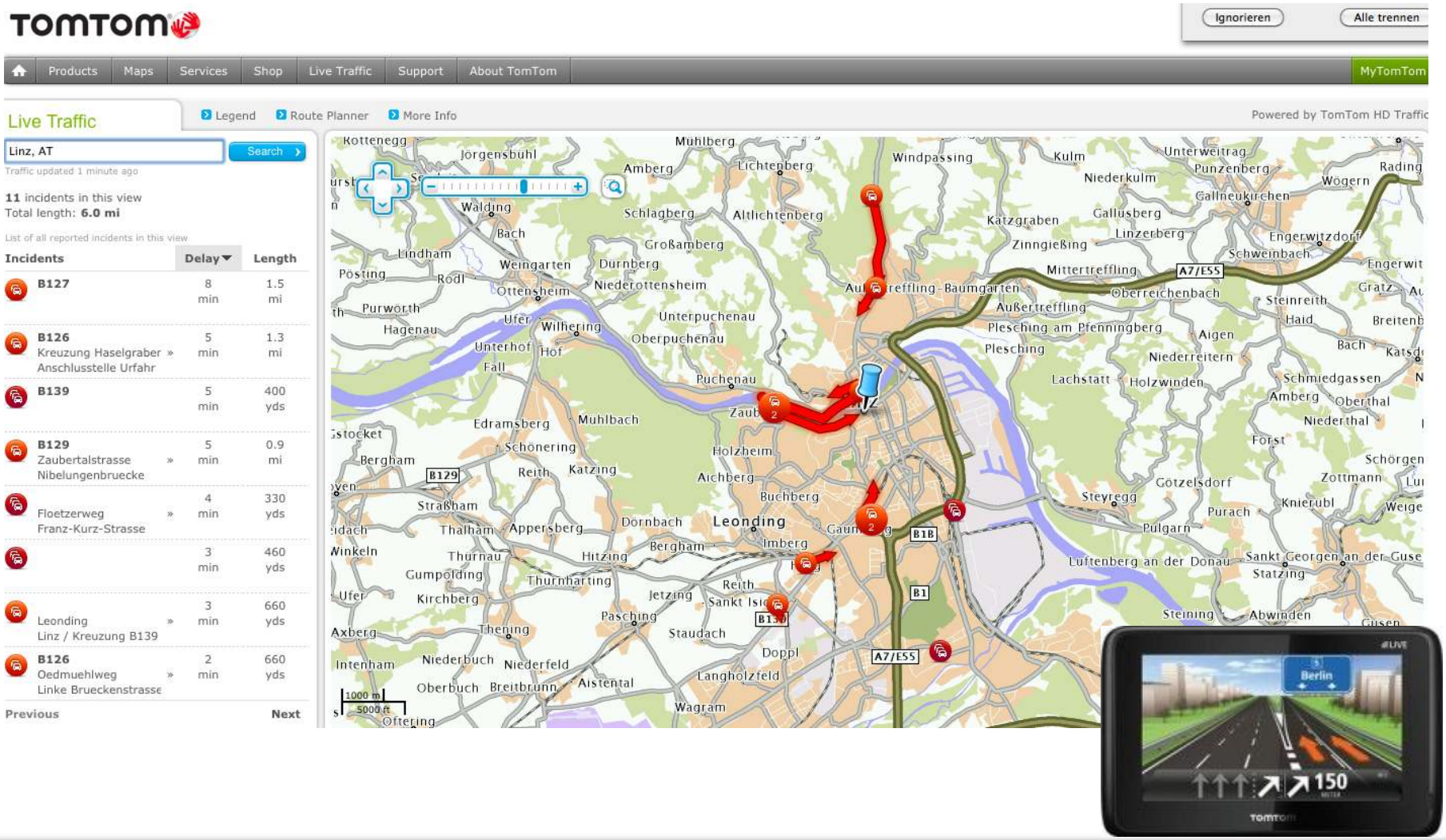
Aims - Traffic Map 2.0

- Road users move as a active source of information inside traffic networks
- Objective of forming a „Community“ to detect traffic conditions – user generated traffic map
- Applications:
  - Traffic situation map on the internet
  - Traffic messages (congestions, delays etc.)
  - Level of Service (LOS) criteria:
    - Travel time
    - Tailback lengths
    - Average waiting time
    - Travel speed



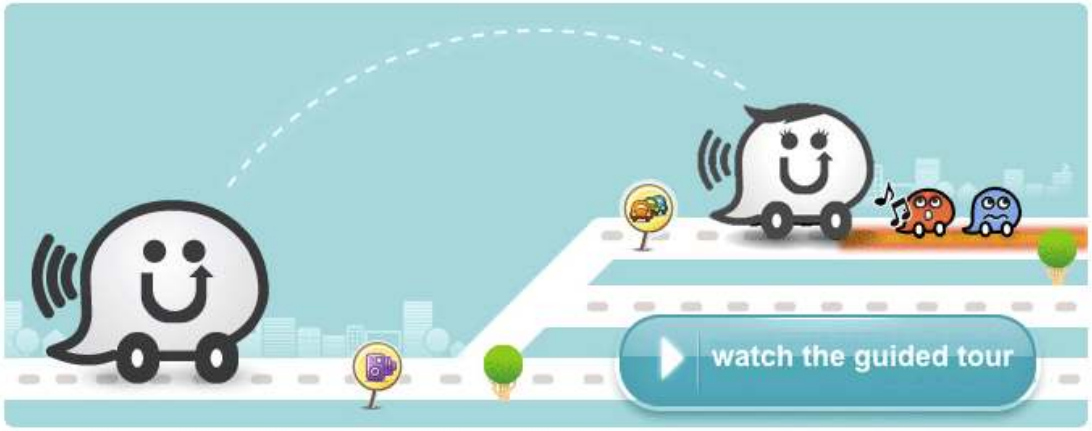


Example: TOMTOM – Live Traffic



Example: Community approach WAZE

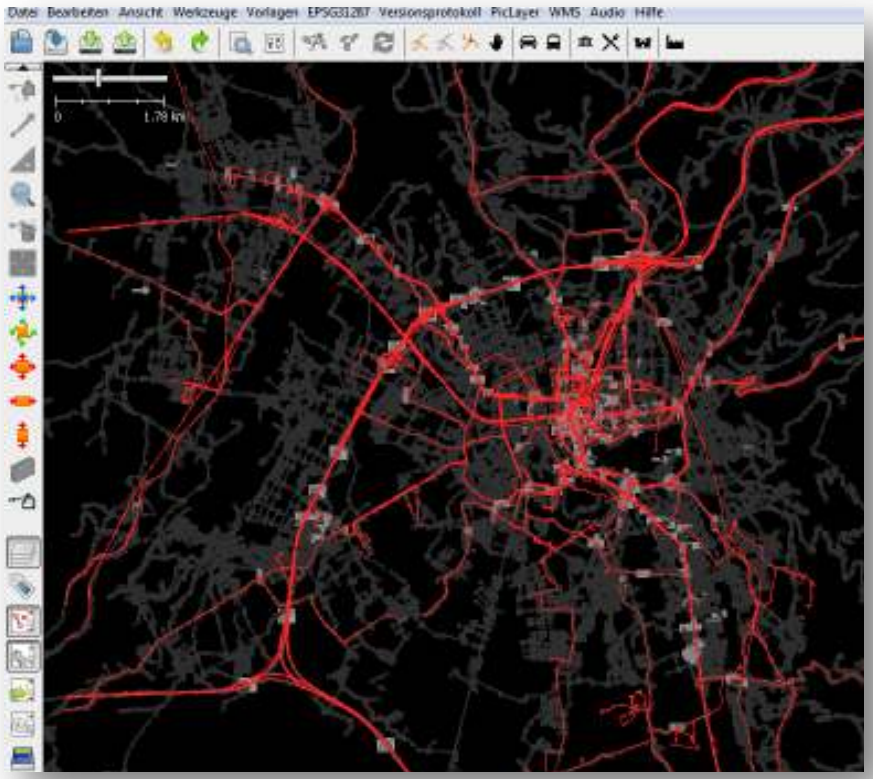




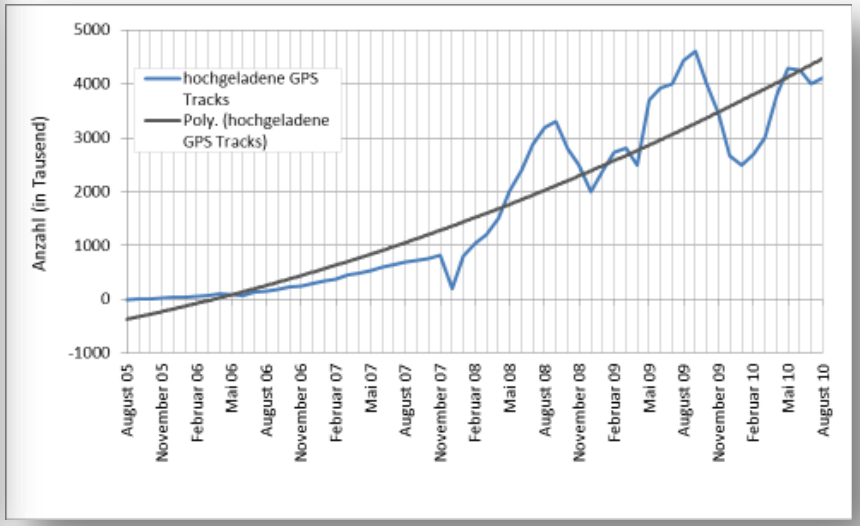


What contribution can the OpenStreetMap project provide concerning traffic detections?

GPS-Tracks OpenStreetMap



JOSM with: OSM Data CC-BY-SA and Contributors



- 1.9 billion GPS coordinates worldwide (August 2010)
- Spatial positioning roads, waterways etc. ➔ use of GPS-Logger
- Traffic flow information can be derived from GPS-Tracks

Method development (2002 – 2010)



Stefan Krampe

„Traffic Detection with Floating-Traveller-Data (FTD) for mobile navigation services“
(Dissertation 2006)
for:
- Deutsche Telekom AG
- Land Hessen (HMWVL)



Enrico Steiger
“Methods of traffic data detection and forecasting from volunteered geographic information for intelligent transportation systems (ITS)”

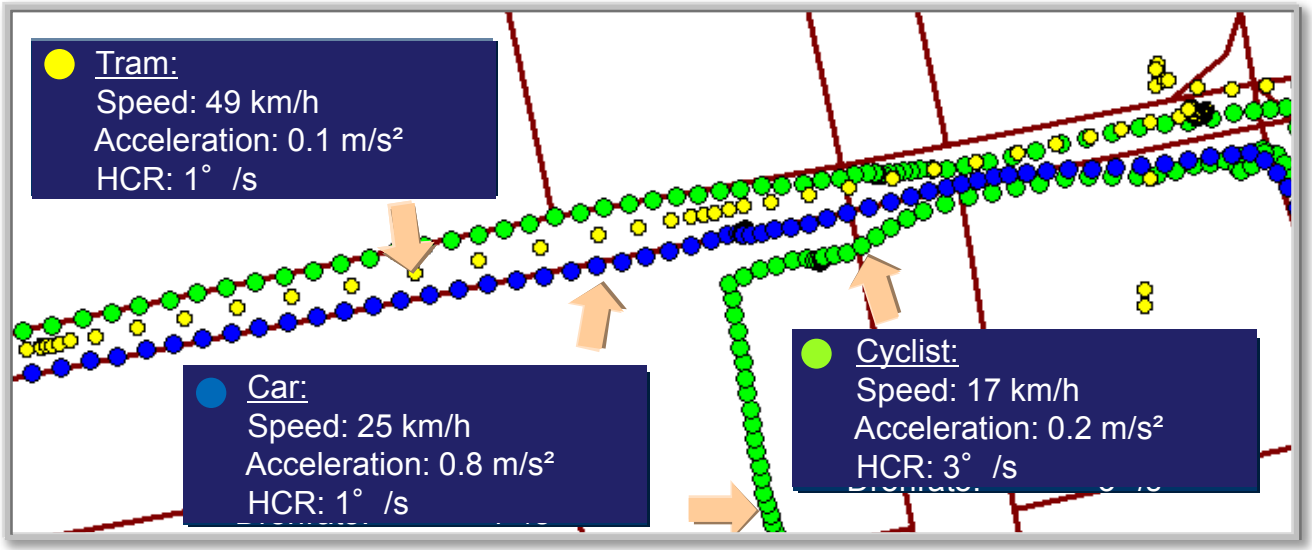
(Diploma Thesis 2010)

Preanalysis of GPS-Tracks

- Evaluation of characteristic GNSS values

$x(t_i), y(t_i)$	Position
$\rho(t_i)$	Movement direction
$\Delta x(t_i)$	Path change in a time intervall
$v(t_i)$	Speed in a time intervall
$b(t_i)$	Acceleration in a time intervall
$\omega\ (t_i)$	Heading change rate (HCR) in a time intervall





Traffic Map 2.0 - Process







GPS-Data

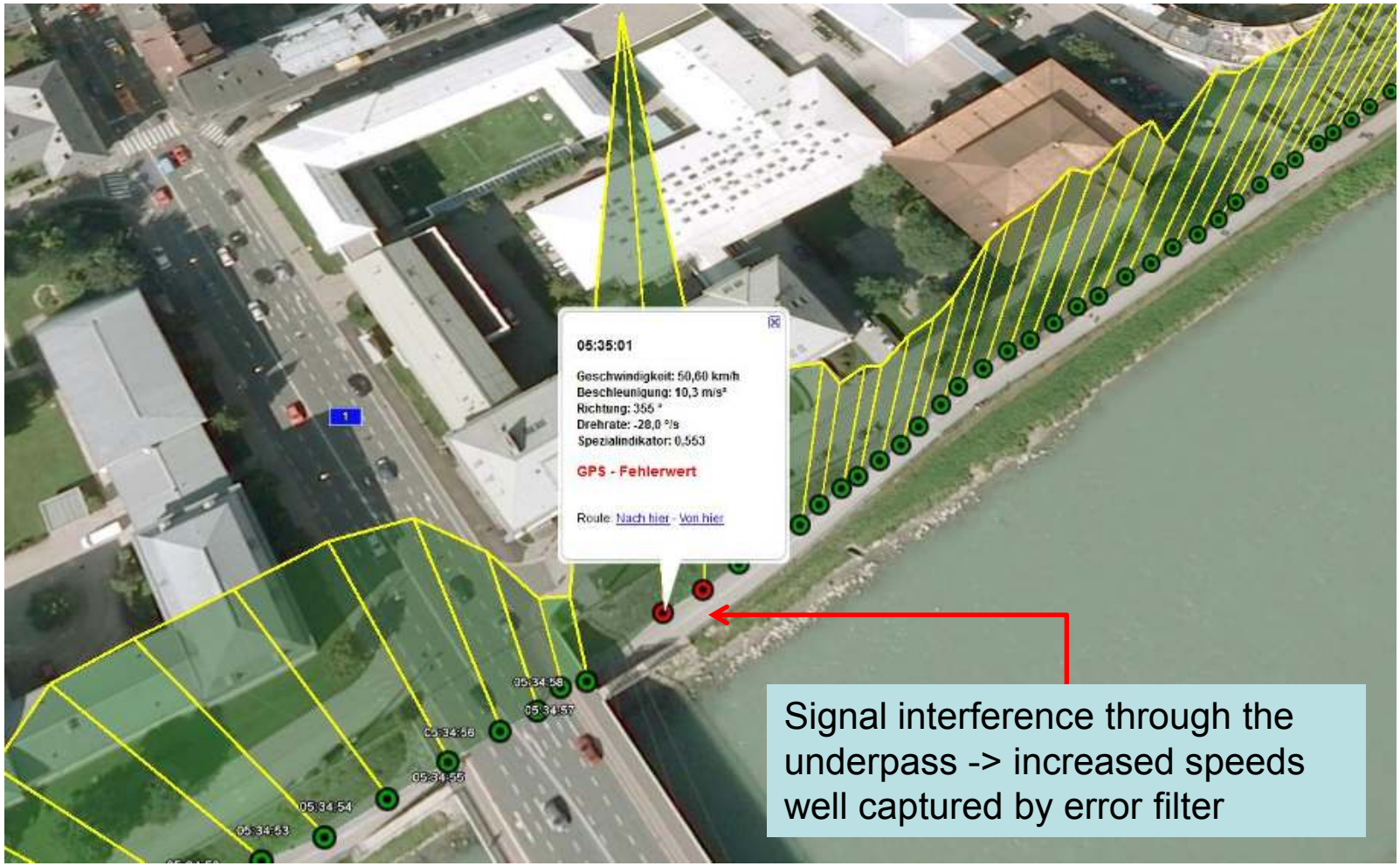
Error detection

Mode of Transport recognition

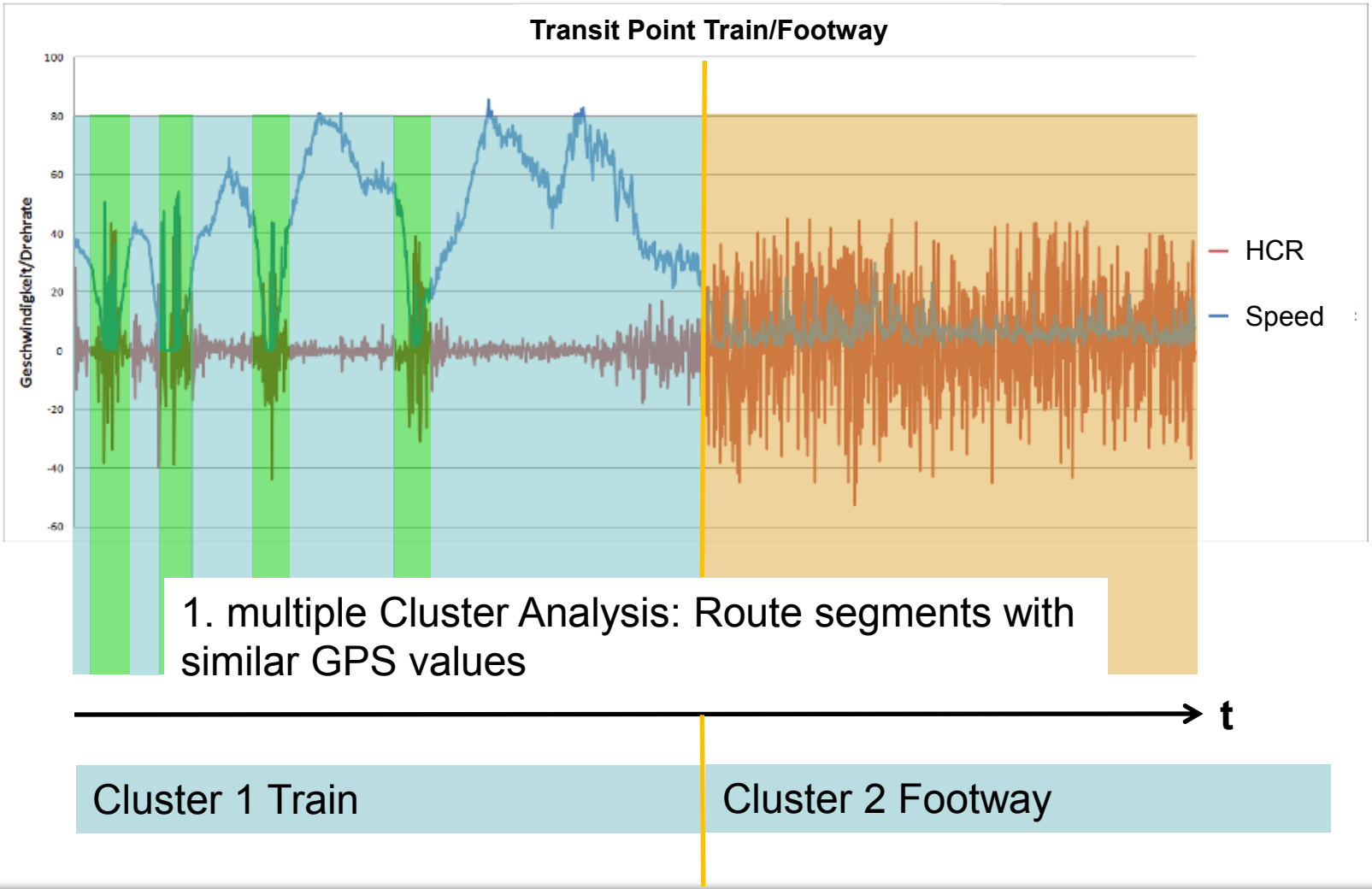
GPS-Track MapMatching on OSM

Traffic Map 2.0

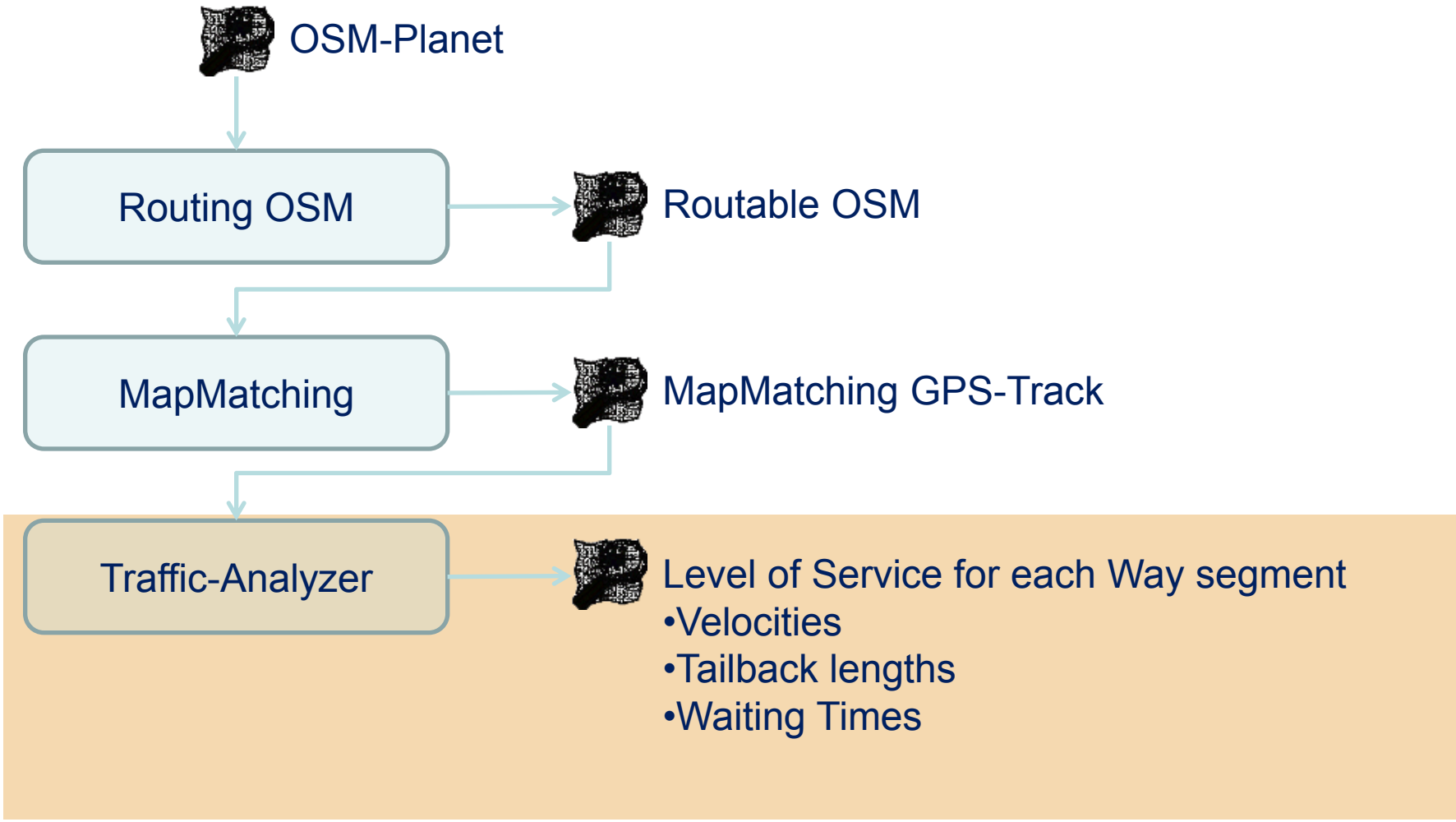
Detecting GPS-Errors



Mode of Transport recognition



Processing OSM GPS Data

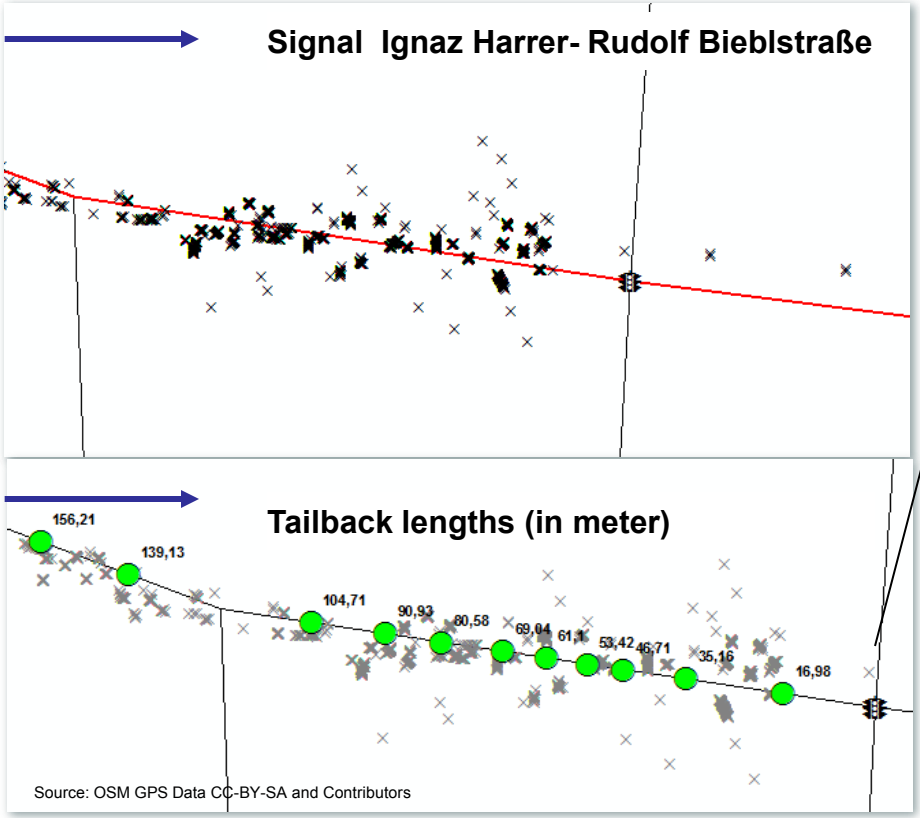


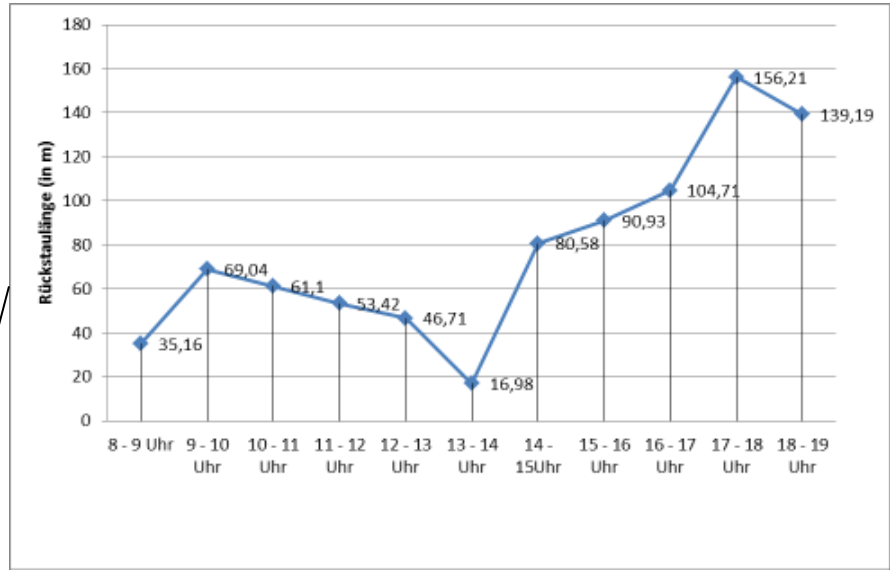
Automatic determination of tailback lengths and average waiting times from GPS Data

- Test area Ignaz-Harrer/ Münchner Bundesstraße

  dailytime change of tailback lenghts at traffic signals

Identification of time-dependent waiting time - cluster centroids





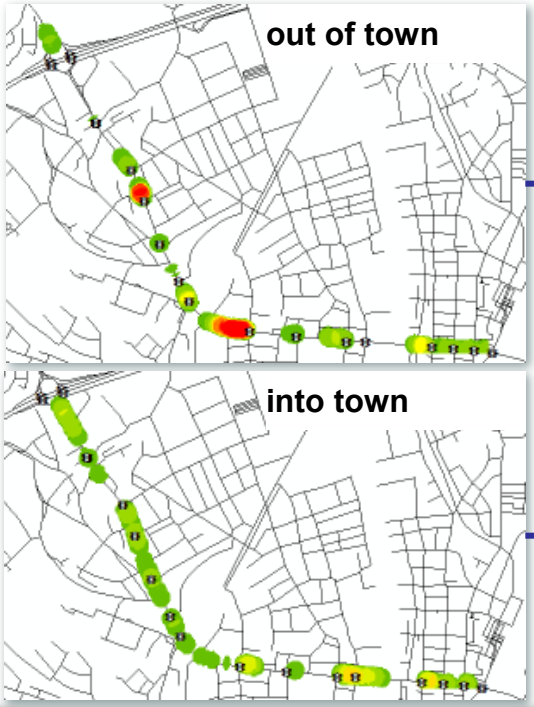
Average tailback length /hour

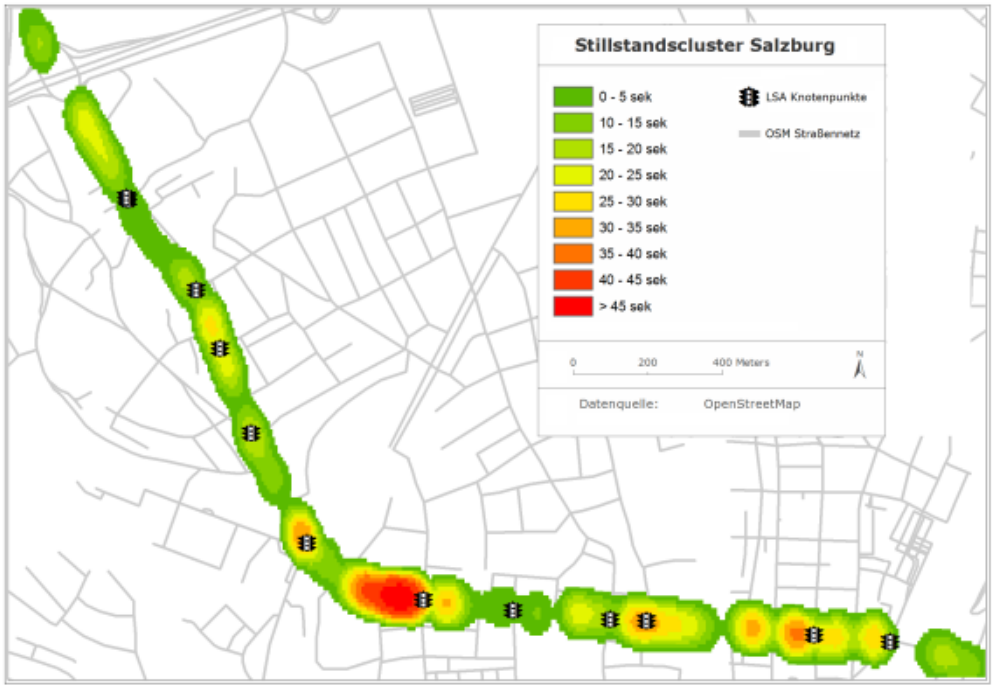
Level of Service
criterion: traffic signal waiting time

- Test area Ignaz-Harrer/ Münchner Bundesstraße

Standstill grids (resolution 10x10 meter grids)

Dependancy frequency distribution of standstills inside road networks and their temporal duration





Source: OSM Data CC-BY-SA and Contributors

Level of Service

- Analyzing speeds

Traffic-Analyzer

GPS-based detection of speeds/ travel times

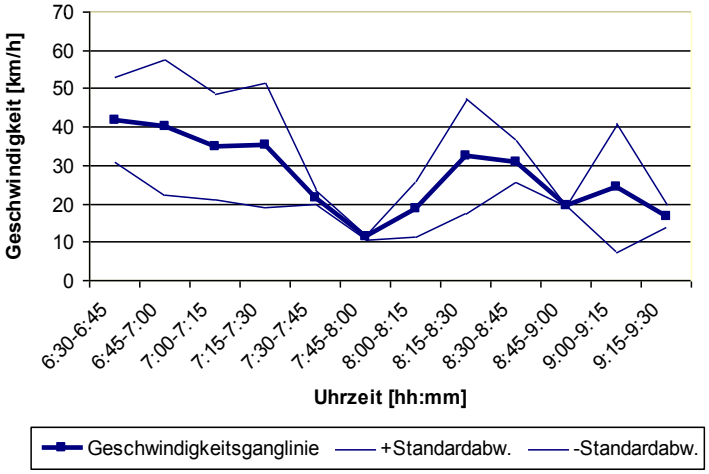
Motorized Traffic

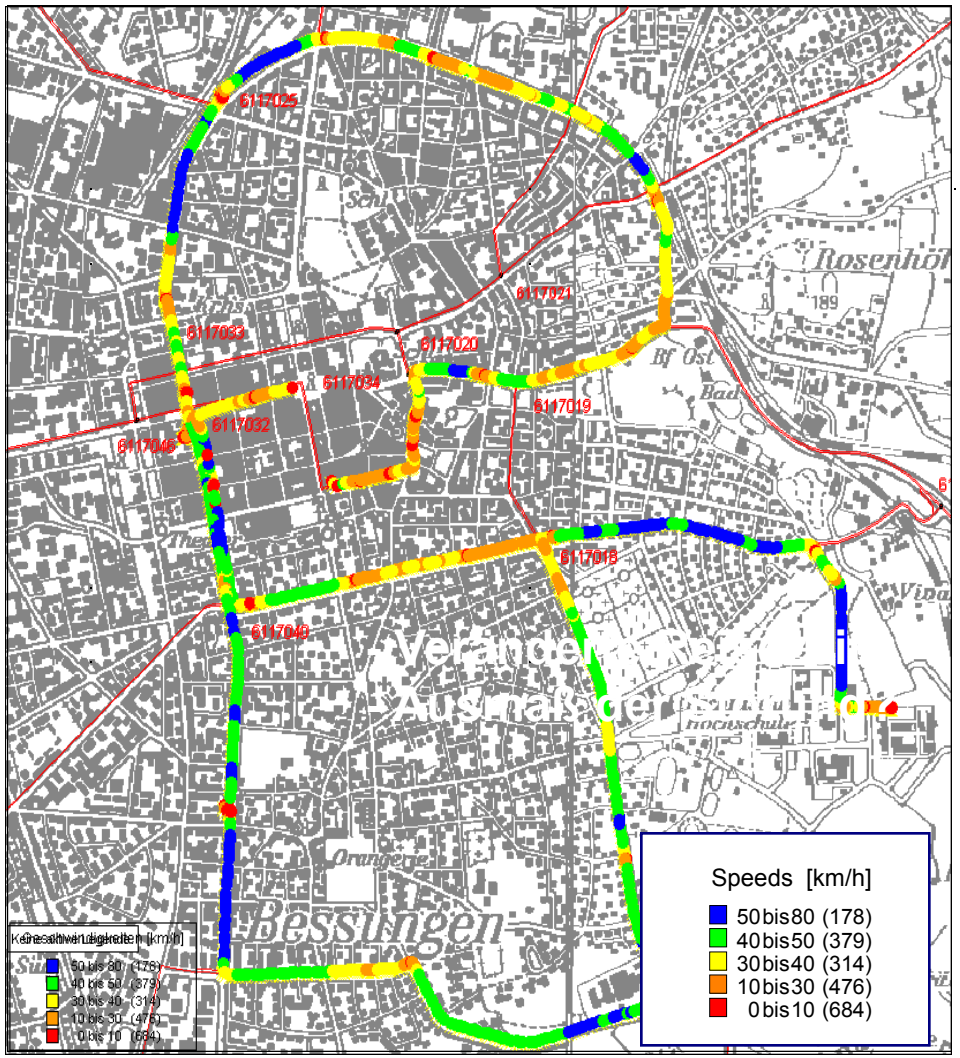


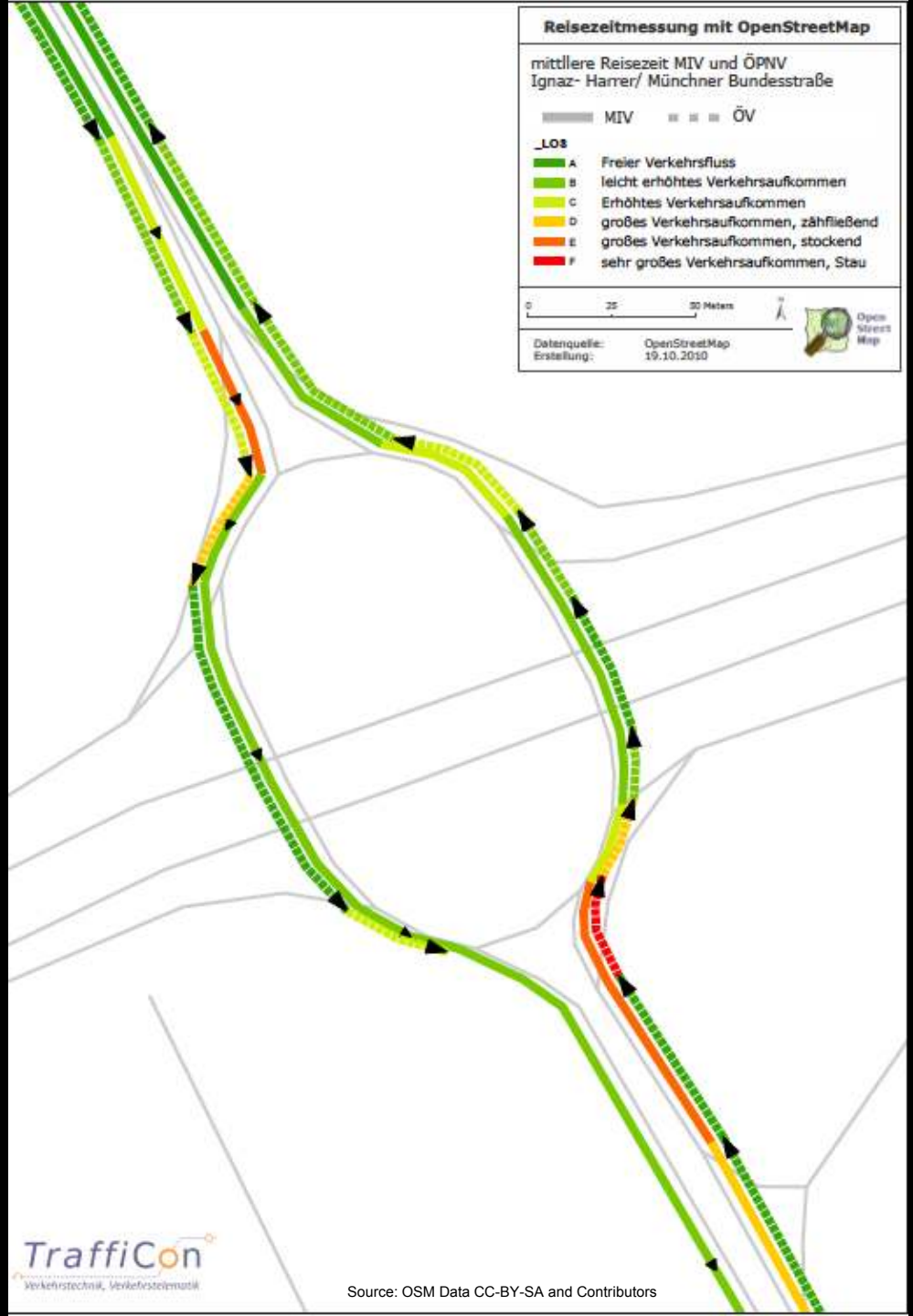
Data:

- Travel time
- Travel speed

Speed Graph and traffic forecast






Reisezeitmessung mit OpenStreetMap
mittllere Reisezeit MIV und ÖPNV
Ignaz- Harrer/ Münchner Bundesstraße
MIV
ÖV
_LOS
A Freier Verkehrsfluss
B leicht erhöhtes Verkehrsaufkommen
C Erhöhtes Verkehrsaufkommen
D großes Verkehrsaufkommen, zähfließend
E großes Verkehrsaufkommen, stockend
F sehr großes Verkehrsaufkommen, Stau
0
25
50 Meters
Datenquelle: OpenStreetMap
Erstellung: 19.10.2010
TraffiCon
Verkehrstechnik, Verkehrstelematik
Source: OSM Data CC-BY-SA and Contributors



Conclusion

- Strong penetration of mobile location technologies within the next years allows an efficient traffic detection
- developed methods and procedures can be a basis for upcoming online traffic models.
- Historical OSM GPS data provides a very good basis for evaluating the quality of transport networks (LOS)

  ➔ Improving traffic planning



Questions?

-

Thank You





TraffiCon – Traffic Consultants GmbH
Enrico Steiger
Schillerstraße 25 - 5020 Salzburg, www.trafficon.eu